End-to-End Sales Data Analysis and Interactive Dashboard Using Excel

Project Objective

To perform an end-to-end sales data analysis using Microsoft Excel by cleaning, transforming, validating, and modeling customer, sales, product, and store datasets. The goal was to generate actionable business insights through pivot tables and an interactive dashboard to support data-driven decision-making.

Dataset Overview

The project uses multiple related datasets to simulate a real-world retail sales environment:

- **Customer Dimension** – customer demographics and loyalty information
- **Sales Fact** – transactional sales data
- **Product Dimension** – product details and inventory
- **Store Dimension** – store location, region, and type

These datasets enable analysis across **customers, products, regions, and time**.

Dataset Structure & Key Columns

Customer_Dim Sheet

Column	Description
Customer_ID	Unique customer identifier
Name	Customer full name
Age	Customer age
Gender	Customer gender

City	City of residence
State	State of residence
Country	Country
Loyalty_Level	Customer loyalty category

Sales_Fact Sheet

Column	Description
Sales_ID	Unique sales transaction ID
Order_Date	Transaction date
Customer_ID	Customer reference
Product_ID	Product reference
Store_ID	Store reference
Quantity	Units sold
Unit_Price	Price per unit
Discount	Discount applied
Total_Amount	Final transaction value
Expected_Total	Calculated transaction value

Product_Dim Sheet

Column	Description
Product_ID	Unique product identifier
Product_Name	Product name
Category	Product category
Sub_Category	Product subcategory
Brand	Brand name
Cost	Product cost
Stock	Inventory availability

Store Dimension

Column Name	Description
Store_ID	Unique identifier assigned to each store
Store_Name	Name of the store
Region	Geographic region of the store (e.g., East, North, South)
City	City where the store is located
Store_Type	Type of store (Flagship, Outlet, or Online)

Customer Dimension – Data Cleaning & Standardization

- ## Customer_ID

- **Data Type:** General
- Standardized the Customer_ID format by converting values such as **Cust** to **C-** using Find & Replace.
- Verified that all Customer_ID values are **unique** with no duplicates.

- ## Name

- **Data Type:** Text
- Cleaned the column using **CLEAN** and **TRIM** to remove extra spaces and non-printable characters.
- Standardized name formatting using **PROPER**.
- Ensured there are **no blank values**.

- ## Age

- **Data Type:** Number
- Removed decimal values and converted the column into whole numbers.
- Ensured age values are consistent and numeric for analysis.

- ### Gender

- **Data Type:** Text
- Data was already well formatted; no cleaning required.

- ### City

- **Data Type:** Text
- Verified text consistency and ensured no formatting issues.

State

- **Data Type:** Text
- Checked for uniform naming and text consistency.

Country

- **Data Type:** Text
- Standardized country names by converting variations such as **"United States of America"** to **"USA"** using Find & Replace.
- Ensured all records contain a single standardized value (**USA**).

- ### Loyalty_Level

- Cleaned the column using **CLEAN** and **TRIM** via a helper column:
  =CLEAN(TRIM(I2))

- Filled missing values with **"Standard"** using:
  =IF(ISBLANK(I2),"Standard",I2)

- Copied and pasted values to replace formulas after cleaning.

Final Validation

- All columns have correct data types.
- No missing or inconsistent values remain.

- Customer Dimension table is **fully standardized and analysis-ready**.

Sales Fact Table

- **Data Type Standardization**
- Converted all ID columns (Sales_ID, Customer_ID, Product_ID, Store_ID) into a consistent standard format.
- Converted Order_Date into **Date format** so that **Year, Month, and Date-based analysis** could be performed.
- Ensured Quantity, Unit_Price, Discount, and Total_Amount were numeric.
- To validate the accuracy of transaction values, a new column called **Expected_Total** was created using the formula:
- Expected_Total = Quantity × Unit_Price × (1 − Discount)
- This column was used to **cross-check whether the given Total_Amount was correct or mismatched**.
- After comparing Total_Amount with Expected_Total:
- Around **12 records showed mismatches or missing values**.
- Issues identified included:
    - Missing **Quantity**
    - Missing **Unit_Price**
    - Missing **Discount**
    - Incorrect **Total_Amount**
- **Reverse Engineering Missing Quantity**
- For records where **Quantity was missing** (e.g., S-8, S-15, S-29, S-34, S-43), Quantity was calculated using reverse engineering:
- Quantity = Total_Amount / (Unit_Price × (1 − Discount))

- Excel formula used:
- =L10 / (G10 * (1 - H10))

	A	B	C	D	E	F	G	H	I	J	K	L
1	Sales_ID	Order_Date	Customer_ID	Product_II	Store_ID	Quantity	Unit_Price	Discou	Payment_Typ	Column2	Expected total	Total_Amoun
2	S-0	7/7/2024	C-1409	Prod-97	Store-3	1	$218.04	0.09	Credit Card	1	$198.42	$198.42
3	S-1	12/25/2023	C-1467	Prod-74	Store-17	3	$104.47	0.27	PayPal	3	$228.79	$228.79
4	S-2	6/3/2024	C-1056	Prod-49	Store-8	4	$914.05	0.25	Credit Card	4	$2,742.15	$2,742.15
5	S-3	11/29/2023	C-1346	Prod-76	Store-8	1	$921.51	0.16	COD	1	$774.07	$774.07
6	S-4	1/6/2024	C-1493	Prod-59	Store-5	3	$745.30	0.18	Credit Card	3	$1,833.44	$1,833.44
7	S-5	7/26/2024	C-1116	Prod-15	Store-7	3	$147.01	0.19	COD	3	$357.23	$357.23
8	S-6	7/12/2025	C-1016	Prod-56	Store-4	2	$785.31	0.15	PayPal	2	$1,335.03	$1,335.03
9	S-7	11/18/2023	C-1152	Prod-70	Store-8	4	$342.39	0.23	PayPal	4	$1,054.56	$1,054.56
10	S-8	05/04/25	C-1187	Prod-99	Store-5	3	$448.25	0.07	PayPal	3	$1,250.62	$1,250.62
11	S-9	7/31/2025	C-1464	Prod-33	Store-0	4	$873.17	0.23	PayPal	4	$2,689.36	$2,689.36
12	S-10	2/13/2024	C-1065	Prod-24	Store-14	1	$278.63	0.07	Credit Card	1	$259.13	$259.13
13	S-11	5/24/2025	C-1320	Prod-39	Store-1	4	$286.92	0.17	Credit Card	3.84	$914.47	$914.47
14	S-12	8/24/2024	C-1237	Prod-88	Store-4	4	$644.58	0.01	Credit Card	4	$2,552.54	$2,552.54
15	S-13	8/6/2025	C-1443	Prod-34	Store-16	2	$316.90	0.30	PayPal	2	$443.66	$443.66
16	S-14	6/21/2025	C-1348	Prod-24	Store-15	4	$403.47	0.29	PayPal	4	$1,145.85	$1,145.85
17	S-15	12/10/24	C-1435	Prod-36	Store-9	3	$871.34	0.04	COD	3	$2,509.46	$2,509.46
18	S-16	9/13/2025	C-1276	Prod-65	Store-19	1	$451.98	0.25	Credit Card	1	$338.99	$338.99

Reverse Engineering Missing Unit Price

- For records where **Unit_Price was missing** (e.g., S-19, S-52, S-268):
- Unit_Price = Total_Amount / (Quantity × (1 − Discount)
- Excel formula used:
  =L21 / (F21 * (1 - H1))

Calculating Missing Discount

- For records where **Discount was missing** (e.g., S-9, S-11, S-16, S-17, S-18, S-26, etc.), Discount was calculated by comparing actual and expected totals:
- Discount = 1 − (Total_Amount / (Quantity × Unit_Price))

- Excel formula used:
  =ROUND(1 - (L60 / K60), 2)

- ## Resolving Mismatched Total_Amount

- After calculating missing values, **Expected_Total became the most reliable value**.
- Wherever mismatches existed, **Total_Amount was updated to match Expected_Total** to ensure consistency and accuracy.

- ## Final Validation & Cleanup

- All calculated helper columns were removed after validation.
- Final checks were performed for:
  - Correct data types
  - No missing values
  - No mismatched totals
- The **Sales Fact table was fully cleaned, standardized, and validated.**

Product Dimension

- Standardized Product_ID format.
- Filled missing Cost values using category-wise averages:

=IF(F3="", AVERAGEIF($C:$C, C3, $F:$F), F3)

- Marked missing Stock values as **Stock Missing** due to unreliable estimation.
- Verified completeness and consistency across all fields.

Store_ID

- **Data Type:** Text
- Verified Store_ID follows a consistent format (Store-#).
- Confirmed all Store_ID values are **unique** with no duplicates.

Store_Name

- **Data Type:** Text
- Cleaned store names using **CLEAN** and **TRIM** to remove extra spaces or hidden characters.
- Ensured consistent naming format across all records.

Region

- **Data Type:** Text
- Verified region values (East, North, South) are consistent.
- No missing or incorrect region values found.

City

- **Data Type:** Text
- Checked for formatting consistency.
- Ensured city names are properly spaced and readable.

Store_Type

- **Data Type:** Text
- Verified store type values (Flagship, Outlet, Online) are consistent.
- No missing values or spelling inconsistencies.

Final Validation

- All columns have correct data types.
- No missing, duplicate, or inconsistent values.
- Store Dimension table is **clean, standardized, and ready for analysis and modeling**.

Data Modeling

- Integrated Loyalty_Level into Sales data using **VLOOKUP**.
- Added Category from Product data to enable category-level analysis.
- Removed redundant Country column since all records belonged to the same country.

Visualization & Dashboard Insights

- KPI Cards:
  - **Total Revenue**
  - **Total Quantity Sold**
  - **Average Discount**
- Regional sales analysis with **payment method breakdown**.
- Sales volume comparison across regions.
- Loyalty-level revenue contribution analysis.
- Dynamic **date slicers** for month and year-wise analysis.

Key Observations & Insights

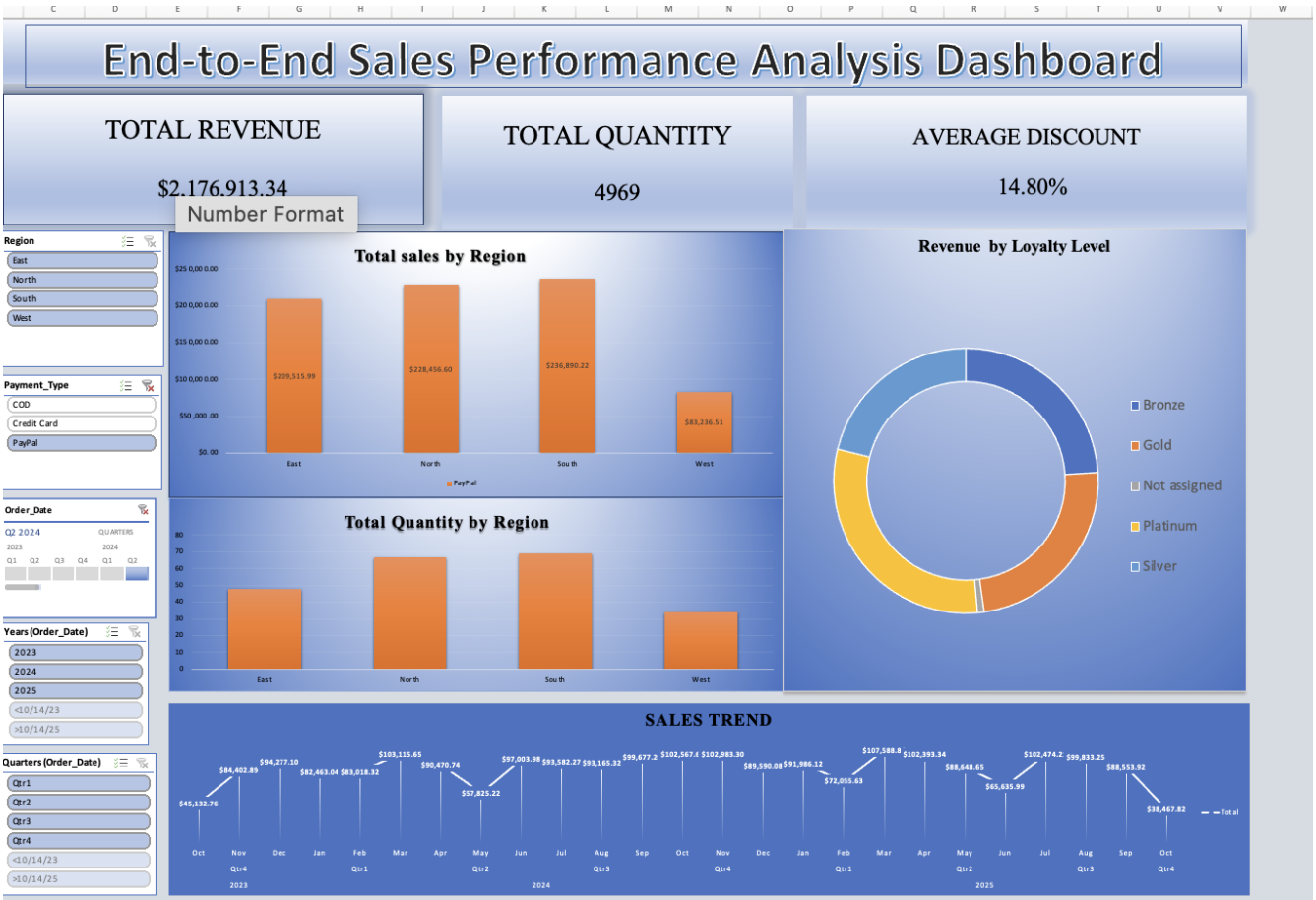
- **North and South regions** generate the highest revenue and sales volume.
- **Platinum and Gold** loyalty customers contribute significantly to total revenue.
- **West region** underperforms in both quantity and revenue.
- Discounts are applied moderately, balancing sales volume and profitability.
- Monthly sales trends indicate **seasonal fluctuations**.

Business Recommendation

- Improve inventory tracking to reduce missing stock data.
- Focus marketing efforts on **high-performing regions and loyal customers**.
- Introduce targeted promotions for underperforming regions.
- Strengthen data validation at the entry level to reduce inconsistencies.

Conclusion

This project demonstrates a complete sales data analysis lifecycle using Microsoft Excel—from data cleaning and transformation to modeling, visualization, and insight generation. The final interactive dashboard delivers clear and actionable insights into sales performance, customer behavior, and regional trends, enabling informed strategic decisions.



Key Insights

- North and South regions generate the highest revenue
- Platinum and Gold customers contribute majority of sales
- West region shows lower volume and revenue